

Dramane Diarra, Ph.D.

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📍 Nairobi, 00200, Kenya

🌐 <https://drdiarra.github.io/DrD-portfolio>



Research Profile

PhD in Electrical Engineering with specialization in Artificial Intelligence, Machine Learning, and next-generation wireless communication systems (5G/6G). Experienced in deep reinforcement learning, optimization, data analysis, and simulation-driven research, with multiple peer-reviewed publications. Strong background in university teaching, interdisciplinary research, and AI-driven system design. Research interests include applied AI at the intersection of cybersecurity, intelligent systems, and digital transformation.

Work Experience

- 04/2023 – 04/2026

📌 **PhD Researcher (Machine Learning and Wireless Systems)**, Pan African University for Basic Science, Technology and Innovation (PAUSTI), Nairobi, Kenya.
Designed and implemented deep reinforcement learning algorithms for energy-efficient resource allocation in large-scale wireless networks (5G/6G). Developed backhaul/fronthaul-aware optimization strategies for heterogeneous systems. Conducted large-scale simulations and performance evaluations, and collaborated with international research teams on AI-driven optimization solutions.
- 09/2023 – 08/2025

📌 **Part-time Lecturer**, Jomo Kenyatta University of Agriculture and Technology (JKUAT), Nairobi, Kenya.
Conducted applied research in signal processing and optimization for communication systems. Supervised engineering projects and implemented simulation-based methodologies. Contributed to curriculum development integrating machine learning and wireless communication technologies.
- 12/2022 – 04/2023

📌 **Telecommunications Engineer**, IMPERIS Sarl, Bamako, Mali.
Designed and deployed optical fiber and microwave communication systems. Performed installation, configuration, and maintenance of telecom infrastructure, including IP-based surveillance systems.
- 09/2022 – 11/2022

📌 **Technical Intern (Network Engineering)**, Orange Mali, Bamako, Mali.
Supported FTTX deployment and configuration for enterprise and residential networks. Contributed to wireless access network implementation and technical documentation processes.
- 12/2022 – 03/2023

📌 **Part-time Lecturer (Telecommunications)**, École Relationnelle d'Informatique et Électronique, Bamako, Mali.
Taught analog and digital communication systems and network planning principles.
- 01/2022 – 08/2022

📌 **Part-time Lecturer (Programming)**, École Universitaire de Technologies et de Gestion (EUTG), Bamako, Mali.
Taught algorithms and Java programming. Designed practical coding exercises and evaluated student performance.
- 02/2021 – 04/2023

📌 **Part-time Lecturer (Electronics and Systems)**, Institut Supérieur des Affaires et de Technologies (ISPATEC), Bamako, Mali.
Delivered courses in electronics, digital systems, Linux, Windows Server, and optical fiber technologies. Supervised laboratory sessions and hands-on projects.

Work Experience (continued)

- 04/2021 – 08/2022  **Part-time Lecturer (Networks and Systems)**, INTEC-Sup, Bamako, Mali.
Taught digital electronics, network management, and Linux system administration with practical configurations.

Education

- 04/2023 – 04/2026  **Ph.D., Pan African University for Basic Science, Technology and Innovation (PAUSTI)** in Electrical Engineering (Telecommunications option).
Thesis title: *Deep Reinforcement Learning for Energy-Efficient Power Control and Allocation in User-Centric Cell-Free Massive MIMO Networks*
- 10/2019 – 07/2021  **M.Sc. Electrical Engineering, University 20 August 1955, Skikda** in Telecommunications Systems.
Thesis title: *Design of a Dual-Band 28/38 GHz MIMO Antenna with High Isolation for 5G Applications Using HFSS Simulation.*
- 10/2016 – 07/2019  **B.Sc. Telecommunications, University 20 August 1955, Skikda.**

Skills

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|------------------|---|
| Languages |  English: fluent (reading, writing, speaking); French: native proficiency. |
| Coding |  Python, R, C++, Java, C#, C, JavaScript, .NET, SQL, L ^A T _E X, PHP, Dart, html/CSS. |
| AI & Data |  Machine Learning, Deep Learning, Reinforcement Learning, Large Language Models (LLMs), Generative AI, Natural Language Processing (NLP), Data Analysis and Visualization, Power BI, Excel. |
| Networking |  TCP/IP Fundamentals, DNS, HTTP, DHCP. |
| Cloud & Systems |  Azure, AWS, Digital Ocean, Linux/Unix, Windows server, Shell scripting. |
| Simulation Tools |  MATLAB, Ansys HFSS, CST, COMSOL, Modelsim, Proteus ISIS. |
| Misc. |  Technical report writing, optimization, academic research, teaching and training. |

Research Publications

Journal Articles

- 1 D. Diarra, H. O. Absaloms, and P. K. Langat, “Deep reinforcement learning-based power control optimization for maximizing energy efficiency in mmwave user-centric cell-free massive mimo,” *IET Communications*, 2026.
- 2 D. Diarra, H. O. Absaloms, and P. K. Langat, “Energy-efficient downlink power allocation in user-centric cell-free massive mimo networks using hybrid deep reinforcement learning algorithm,” *IEEE Transactions on Communications*, 2026.
- 3 D. Diarra, H. O. Absaloms, and P. K. Langat, “Multi-agent deep reinforcement learning-based power control for energy-efficient user-centric cell-free massive mimo,” *Physical Communication*, 2026.
- 4 D. Diarra, H. O. Absaloms, and P. K. Langat, “Optimizing uplink power control for energy efficiency in mmwave user-centric cell-free massive mimo with deep reinforcement learning,” *Journal of Engineering and Applied Science*, vol. 73, 2026.
- 5 D. Diarra, H. O. Absaloms, and P. K. Langat, “Deep reinforcement learning-based power allocation for maximizing energy efficiency in mmwave cell-free massive mimo,” *International Journal of Intelligent Engineering and Systems*, vol. 18, 2025.

Miscellaneous Experience

Awards and Achievements

- 2023  **Pan African University (PAUSTI) Scholarship.**
- 2016  **Algeria–Mali Cooperation Scholarship (Excellence Award).**

Certification

- 2025  **Learning AI Through Visualization.** Awarded by Columbia University.
- 2023  **English Language Level – Intermediate.** Awarded by Pan African University Institute for Basic Sciences, Technology, and Innovation.
- 2020  **English Language Certification (B2 – Independent User).** Awarded by Intensive Languages Teaching Centre at University of Skikda, Algeria.

References

Available on Request